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Team Larpers Pilot Plan

Route2Zero Six-Month City Pilot

City pilot question

Can Route2Zero produce a city-owned, evidence-backed e-jeepney Phase-1 portfolio whose climate, equity, infrastructure, operator, and uncertainty assumptions have been tested against real local evidence?

PILOT PURPOSE

Convert Route2Zero 2.0 from an evidence-based prototype into a locally validated planning workflow that cities can inspect, challenge, refresh, and own.

PILOT SCOPE

- Target up to three partner LGUs, subject to city pairing and pilot agreement.
- Validate a manageable corridor sample, one flagship corridor, and a broader Phase-1 shortlist.
- Engage relevant operators/cooperatives, a utility or energy counterpart, and rider/community representatives where appropriate.
- Test ML service estimates, climate assumptions, route geometry, equity evidence, charging evidence, operator readiness, policy scenarios, robustness, and portfolio constraints.

SIX-MONTH OUTCOME

Outcome area	Deliverable
Decision	City-owned Phase-1 portfolio or shortlist with named next actions.
Evidence	Validated route cards, source registry, validation ledgers, and unresolved evidence owners.
Models	Calibrated service-intensity model, climate scenarios, model card, and evidence-confidence rules.
Operations	Scenario library, decision pack, refresh/retraining runbook, and reproducibility handover.

PILOT PATHWAY

PROTOTYPE	VALIDATE	CALIBRATE	CO-DESIGN	HANDOVER

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Six months, three evidence gates

Timeline	Activity / Milestone	Core activity	Outcome
Month 1	Baseline + model protocol	Confirm decision owners and route sample; freeze source registry/build; approve validation, ML evaluation, privacy, and climate protocols.	Pilot charter; baseline build; route sample; validation protocol; risk register.
Month 2	Data refresh + ground truth	Desk/field-check routes; observe service; validate geometry; collect operator, charging/depot, and local equity/accessibility evidence.	Updated validation, operator, and charging ledgers; ground-truth sample; source refresh.
Gate 1	Evidence readiness review	Accept evidence plan and model-validation protocol; confirm minimum data coverage; resolve governance blockers.	Approved evidence gate or documented corrective actions.
Month 3	Model + route validation	Compare ML estimates with refreshed observations; recalculate metrics; review anomalies; tune geometry/evidence rules and equity variables.	Pilot model evaluation; model-card update; revised evidence grades; validated flagship case.
Month 4	Climate, charging + operator pre-feasibility	Calibrate diesel/electric assumptions; validate energy scenarios; request utility evidence; assess sites, operators, and verified constraints.	Calibrated climate scenario; charging/operator cards; optimizer-ready constraints.
Gate 2	Quality + route-set review	Accept pilot route set and model/evidence quality; hold routes with unresolved critical issues or mark them evidence-limited.	Approved route set with explicit holds and evidence limits.
Month 5	Scenario co-design + human factors	Run policy workshops; review sensitivity; agree portfolio constraints; test Priority vs Evidence vs Stability; review assistant and accessibility.	Approved scenarios; draft portfolio; usability and governance findings.
Month 6	Decision pack + handover	Finalize portfolio, evidence queue, model card, source registry, refresh process, issue log, analyst training, and task ownership.	Accepted decision pack; final build; reproducibility package; city handover.
Gate 3	Acceptance	City decision pack and reproducibility handover accepted by named decision owners.	Go-forward decision and accountable next-step owners.

Baseline traceability: build r2z-16690ccbe328 | scenario scn-c46e1d86c1 | service model service-v1-266e0b1d

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The pilot succeeds when the evidence and decision improve

SUCCESS METRICS

Measure	Pilot success evidence
Coverage	% pilot routes ground-checked; % shortlisted routes with complete evidence cards; critical gaps resolved.
Model	MAE/RMSE vs refreshed observations, improvement over baseline, and documented route-level residual review.
Decision quality	Evidence-grade improvement; ranking/portfolio stability before vs after validation; routes moved from evidence-limited to decision-ready.
Adoption	LGU/operator/utility interactions; planner task completion; understanding of Priority, Evidence, and Stability.
Handover	One accepted city decision pack and one completed reproducibility/refresh handover.

VALIDATION METHODS

Field observations; operator interviews; LGU review; utility evidence requests; data reconciliation; grouped holdout testing; scenario workshops; usability/accessibility testing; stakeholder review of the flagship corridor.

RESPONSIBLE AI CONTROLS

- No resident-level personal data are required for core analytics; stakeholder data are consent-based.
- No hidden LLM score changes and no automatic funding or procurement decisions.
- No individual poverty or informal-status inference from population density.
- Stakeholders can challenge or replace evidence; an AI-disabled deterministic fallback remains available.

RISK REGISTER

Risk	Control / response
Historic route data	Use current-validation ledger and field checks; retain historic-only status until verified.
Weak ML performance	Prefer observed evidence; mark model experimental; compare with baseline.
Utility capacity unavailable	Show mapped proximity only; keep capacity NOT VERIFIED; request evidence.
Operator data unavailable	Keep neutral prior explicit and raise validation priority.
Sparse geometry / equity data	Use reliability/evidence grades; validate traces; avoid unsupported settlement claims.
AI/API unavailable	Serve structured deterministic fallback; scores, scenarios, and portfolio are unaffected.
Constraint conflict	Show infeasibility and binding constraints; never fabricate a portfolio.

Named owners from data collection to city handover

TEAM STRUCTURE

Name	Role	Relevant background / experience / skills for this role
John Marwin Ebona	Product & Pilot Lead	City coordination, decision question, gates, integration, decision pack, handover.
Prince Marl	Data, ML & Pipeline Lead	Source registry, feature store, service model, metrics, build manifest, refresh.
Joaquin Sarmiento	Transport & Field Validation Lead	Route sampling, observations, status reconciliation, assumptions, operator workflow.
Isaac Marcus	Engineering, Optimization & Deployment Lead	Dashboard, scenarios, optimizer, Netlify, Mapbox, reliability, fallback behavior.
Andrei Dela Cruz	Policy, Governance & Responsible AI Lead	Policy controls, decision rights, workshops, risk/change control, responsible AI.
Russel Mendez	Climate, Research & Evidence Lead	Climate/energy method, source validation, confidence framework, documentation.
Tj Moreno	Stakeholder & Partnership Lead	LGU, operator, rider/community, utility and finance coordination and consent.
JM Palaganas	QA, Testing & Documentation Lead	Testing, accessibility, artifact QA, claim-output checks, issue tracking.
Carl Nueva	Geospatial & Infrastructure Analyst	Geometry review, spatial joins, equity layers, charging/power evidence, map QA.
Curt Mathew Garcia	Frontend & UX Developer	Responsive Route Lens, portfolio UX, walkthrough mode, usability, accessibility.
Marck Morgado	Field Validation & Logistics Coordinator	Field schedules, forms, route-check records, evidence handoff, logistics.

PARTNER RESPONSIBILITIES

Partner	Pilot responsibility
LGU	Define decision context, validate evidence, join scenario workshops, review outputs, accept handover.
Operator / cooperative	Validate operations, fleet/depot constraints, readiness, and feasibility.
Utility / energy	Review charging assumptions and flag formal capacity/interconnection studies.
Riders / community	Validate access concerns and challenge equity assumptions where relevant.

HANDOVER AND SCALE

Handover: final build, model card, source registry, validation ledgers, scenario library, decision pack, issue register, refresh/retraining runbook, governance notes, and a city owner for each unresolved issue. Scale requires local adapters, recalibration, and validation; the Metro Manila model is not copied unchanged.